

Assignment-2

1. Impact of Screen Time on Sleep Quality (Correlation)

Generate 250 synthetic records with screen_time, sleep_duration, weekend flag. Compute Pearson, scatter plot, and interpret.

2. Predicting House Rent Using Building Features (Multiple Regression)

Generate 600 records with area, rooms, distance from metro, zone, and rent. Fit regression and evaluate.

3. Effect of Discounts on Online Sales (Regression)

Generate 400 products with discount and units_sold (Poisson). Fit linear and quadratic models.

4. Hypothesis Test on Gender & Learning Mode (Chi-square)

Generate 400 respondents with gender and online/offline preference. Perform chi-square test.

5. Temperature vs Electricity Consumption (Regression + Seasonality)

Generate 365 daily temperature and electricity usage. Fit regression and identify patterns.

7. Salary Prediction (Multicollinearity)

Generate 300 records with experience, certifications, salary. Fit regression.

9. Advertising Budget vs App Downloads (Polynomial Regression)

Generate 48 monthly points with ad_budget and downloads. Fit polynomial model.